

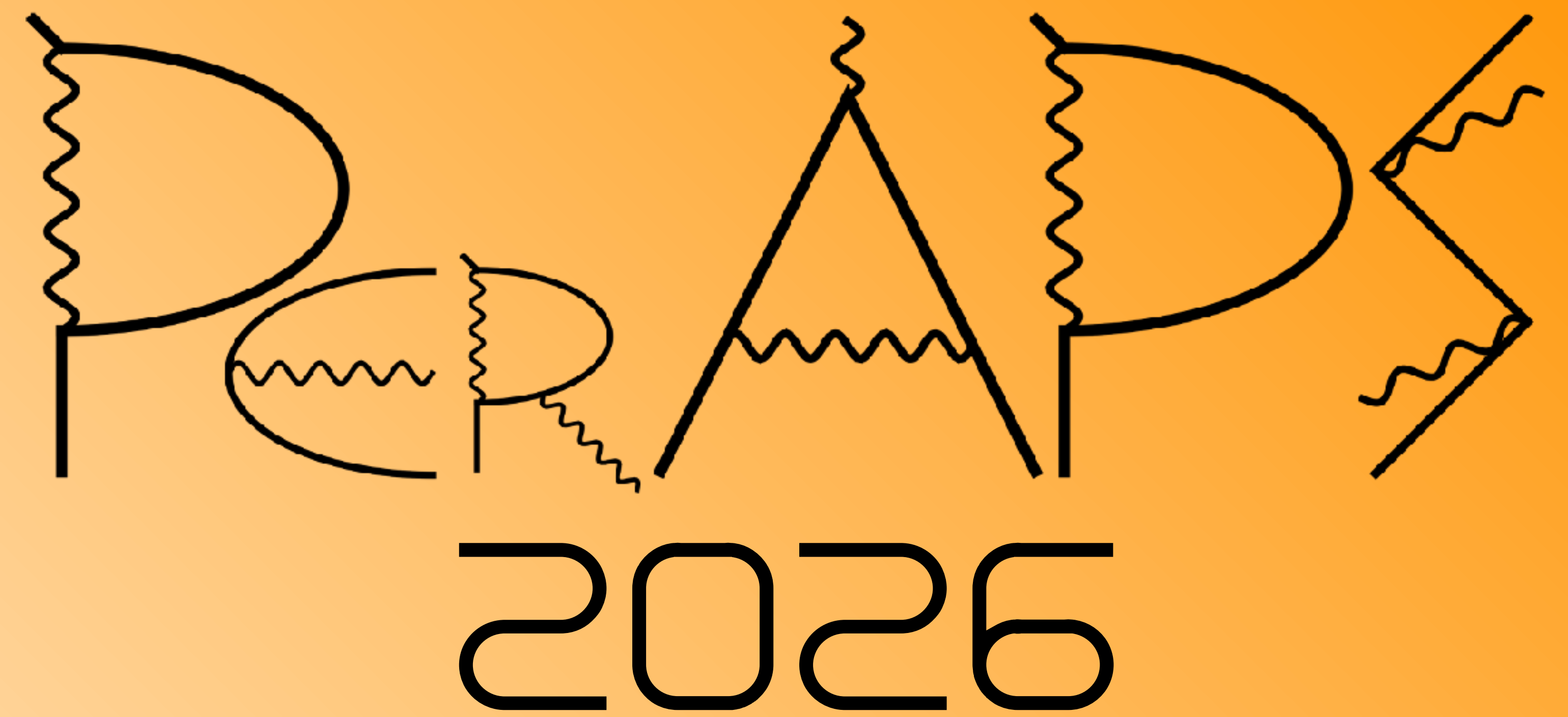
Martedì
13 ottobre
Aula B, ore 14:30

Black Holes as Laboratories for Fundamental Physics: Superradiant Instabilities

Spinning black holes can develop superradiant instabilities in the presence of ultralight bosonic fields. Through this process, a macroscopic bosonic condensate can form around the black hole, producing striking observational signatures, including detectable gravitational waves. Superradiance therefore offers a unique way to search for new light particles, such as axions and dark photons, complementary to conventional laboratory experiments.

However, much of our current understanding of black-hole superradiance relies on treating these fields as non-interacting. This is an idealized picture: realistic bosonic fields generally come with interactions that can significantly alter the evolution of the instability. Axions are a prime example, being both well-motivated targets for superradiant searches and intrinsically self-interacting.

In this talk, I will introduce the physics of black-hole superradiance and show how interactions can qualitatively change both the evolution of the instability and its observable signatures. In particular, I will discuss how they can suppress some of the phenomena predicted in the non-interacting picture, while also giving rise to entirely new signals. These effects are therefore crucial for understanding what we can realistically hope to observe, and for assessing the potential of black holes as laboratories for fundamental physics.



Enrico Cannizzaro

